

Understanding Variational Autoencoders from a mathematical view

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What I cannot create, I do not understand.

Outline

- 1 Introduction: The Problem of Generative Modeling
- 2 Variational Inference: The Foundation
- 3 VAE Variants and Extensions
- 4 Advanced Topics
- 5 Practical Considerations
- 6 Conclusion

The Low-Dimensional Manifold Hypothesis

Key observation: Natural images lie on a low-dimensional manifold, not in the full high-dimensional space.

- Full space: $1024 \times 1024 \times 3 = 3,145,728$ dimensions
- Actual manifold: Much lower dimensional (often < 100 dimensions)
- Most of the high-dimensional space is empty or contains unrealistic images

Central question: How can we embed these features into a compact latent space?

- Capture the essential structure of the data
- Enable efficient representation and generation
- Learn meaningful features for downstream tasks

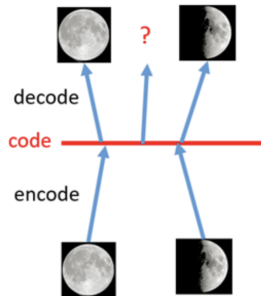
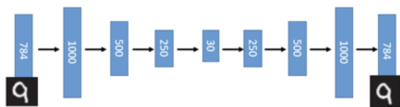
Autoencoders: A Direct Approach

Basic idea: Use an encoder-decoder architecture to learn compact representations

Autoencoder (AE) Architecture

- Encoder: $x \rightarrow z$ (compression)
- Decoder: $z \rightarrow \hat{x}$ (reconstruction)
- Objective: Minimize reconstruction error $\|x - \hat{x}\|^2$

Problem: Interpretation problems in the latent space



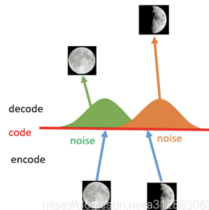
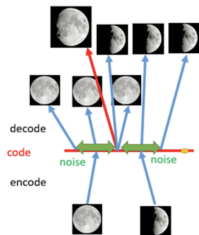
From Deterministic to Probabilistic: Variational Autoencoders

Key insight: Model encoder and decoder as **random variables**

- Encoder: $q_{\phi}(z|x)$ - maps data variable x to latent variable z
- Decoder: $p_{\theta}(x|z)$ - reconstructs data from latent variable
- Prior: $p(z)$ - regularizes the latent space

Benefits:

- **Natural interpolation** in the latent space
- Smooth transitions between data points
- Well-structured, continuous latent representation



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Latent Variable Generative Models

Goal. Learn a probabilistic model that can:

- Assign high probability to data: $x \sim p_{\text{data}}(x)$,
- Generate realistic samples, item Provide useful latent representations z .

Latent variable model.

$$p_{\theta}(x, z) = p_{\theta}(x | z) p(z),$$
$$p_{\theta}(x) = \int p_{\theta}(x, z) dz = \int p_{\theta}(x | z) p(z) dz.$$

Maximum likelihood objective.

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{x \sim p_{\text{data}}} [\log p_{\theta}(x)].$$

In practice, the expectation is replaced by the empirical average over the dataset.

Why the Posterior is a Problem?

We would like to optimize the log-likelihood

$$\log p_{\theta}(x) = \log \int p_{\theta}(x, z) dz.$$

Its gradient (assuming we can interchange ∇ and $\int$):

$$\begin{aligned}\nabla_{\theta} \log p_{\theta}(x) &= \frac{1}{p_{\theta}(x)} \int \nabla_{\theta} p_{\theta}(x, z) dz \\ &= \int \frac{p_{\theta}(x, z)}{p_{\theta}(x)} \nabla_{\theta} \log p_{\theta}(x, z) dz \\ &= \mathbb{E}_{p_{\theta}(z|x)} [\nabla_{\theta} \log p_{\theta}(x, z)].\end{aligned}$$

Key issue: this expectation is taken under the posterior

$$p_{\theta}(z | x) = \frac{p_{\theta}(x, z)}{p_{\theta}(x)},$$

which is typically intractable in deep models: no closed form, difficult to sample from directly.

Variational Inference and the ELBO (Pointwise in x)

Idea: introduce a tractable approximate posterior $q_\phi(z | x)$ and derive a lower bound on $\log p_\theta(x)$. Rewrite:

$$\begin{aligned}\log p_\theta(x) &= \log \int p_\theta(x, z) dz \\ &= \log \int q_\phi(z | x) \frac{p_\theta(x, z)}{q_\phi(z | x)} dz \\ &= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z | x)} \right].\end{aligned}$$

Apply Jensen's inequality ($\log$ is concave, so that $\mathbb{E}_p[\log X] \leq \log \mathbb{E}_p[X]$):

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z | x)} \right] := \mathcal{L}(\theta, \phi; x).$$

$\mathcal{L}(\theta, \phi; x)$ is the Evidence Lower Bound (ELBO). We maximize it w.r.t. both θ and ϕ .

What is the gap?

Start from the KL divergence between $q_\phi(z | x)$ and the true posterior $p_\theta(z | x)$:

$$\begin{aligned}\text{KL}(q_\phi(z | x) \| p_\theta(z | x)) &= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z | x)}{p_\theta(z | x)} \right] \\ &= \mathbb{E}_{q_\phi(z|x)} [\log q_\phi(z | x) - \log p_\theta(z | x)].\end{aligned}$$

Use Bayes' rule $p_\theta(z | x) = \frac{p_\theta(x, z)}{p_\theta(x)}$,

$$\Rightarrow \text{KL} = \mathbb{E}_{q_\phi(z|x)} [\log q_\phi(z | x) - \log p_\theta(x, z) + \log p_\theta(x)].$$

Separate the terms (note $\log p_\theta(x)$ does not depend on z):

$$\text{KL} = \mathbb{E}_{q_\phi} [\log q_\phi(z | x) - \log p_\theta(x, z)] + \log p_\theta(x).$$

Define the ELBO

$$\mathcal{L}(\theta, \phi; x) := \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x, z) - \log q_\phi(z | x)],$$

so that

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + \text{KL}(q_\phi(z | x) \| p_\theta(z | x)).$$

Summarize VAE from the generative perspective

We have $p_{\text{data}}(x)$ and we want to learn a model $p_{\theta}(x)$ that can generate realistic samples from the easily sampled latent variable z .

- We can use the maximum likelihood objective to learn the model $p_{\theta}(x)$.

$$\nabla_{\theta} \log p_{\theta}(x) = \mathbb{E}_{p_{\theta}(z|x)}[\nabla_{\theta} \log p_{\theta}(x|z)]$$

- However, the gradient is intractable because the integral is over the conditional distribution $p_{\theta}(z|x)$ and we do not know the posterior $p_{\theta}(z|x)$
→ We replaced the posterior $p_{\theta}(z|x)$ with a neural network $q_{\phi}(z|x)$.
- Compute the KL divergence between $q_{\phi}(z|x)$ and $p_{\theta}(z|x)$ gives the gap:

$$\text{KL}(q_{\phi}(z|x) || p_{\theta}(z|x)) = \log p_{\theta}(x) - \text{ELBO}(x; \theta, \phi)$$

Remark

Why did we not directly optimize $p_{\theta}(x)$?

Two Views of the ELBO

1. Joint-form ELBO and true posterior gap.

$$\begin{aligned}\mathcal{L}(\theta, \phi; x) &= \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x, z) - \log q_{\phi}(z | x)], \\ \log p_{\theta}(x) &= \mathcal{L}(\theta, \phi; x) + \text{KL}(q_{\phi}(z | x) \| p_{\theta}(z | x)).\end{aligned}$$

Thus for fixed θ :

$$\max_{\phi} \mathcal{L}(\theta, \phi; x) \iff \min_{\phi} \text{KL}(q_{\phi}(z | x) \| p_{\theta}(z | x)).$$

2. Reconstruction-regularization form.

Using $p_{\theta}(x, z) = p_{\theta}(x | z)p(z)$:

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x | z)] - \text{KL}(q_{\phi}(z | x) \| p(z)).$$

- First term: reconstruction (log-likelihood under decoder).
- Second term: regularization toward prior $p(z)$.

The VAE Architecture

We parameterize both the generative model and the approximate posterior with neural nets.

Prior over latents.

$$p(z) = \mathcal{N}(0, I).$$

Decoder (generative model).

$$p_{\theta}(x | z)$$

- Example (continuous): $p_{\theta}(x | z) = \mathcal{N}(\mu_{\theta}(z), \sigma^2 I)$.
- Example (binary): $p_{\theta}(x | z) = \text{Bernoulli}(\text{sigmoid}(f_{\theta}(z)))$.

Encoder (approximate posterior).

$$q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x))),$$

where $\mu_{\phi}(x)$ and $\log \sigma_{\phi}^2(x)$ are outputs of a neural network.

Training objective (per-sample).

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x | z)] - \text{KL}(q_{\phi}(z | x) \| p(z)).$$

Reparameterization Trick

To optimize the ELBO with SGD, we need gradients through the expectation

$$\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x | z)].$$

Directly sampling $z \sim q_{\phi}(z | x)$ makes the gradient w.r.t. ϕ hard to propagate.

Reparameterization trick:

$$\begin{aligned} z &\sim q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x))) \\ \implies z &= \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \epsilon, \\ \epsilon &\sim \mathcal{N}(0, I). \end{aligned}$$

Now the expectation becomes

$$\mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\log p_{\theta}(x | \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \epsilon)],$$

which is differentiable w.r.t. both θ and ϕ using standard backprop.

Dataset Objective and Training

Population objective:

$$\max_{\theta, \phi} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\mathcal{L}(\theta, \phi; \mathbf{x})].$$

Approximate with empirical average over dataset $\{\mathbf{x}_i\}_{i=1}^N$:

$$\max_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\theta, \phi; \mathbf{x}_i).$$

Mini-batch SGD:

- Sample a mini-batch $\{\mathbf{x}_i\}_{i \in \mathcal{B}}$,
- For each $\mathbf{x}_i$, sample $\epsilon^{(i)} \sim \mathcal{N}(0, I)$,
- Compute $\mathbf{z}^{(i)} = \mu_{\phi}(\mathbf{x}_i) + \sigma_{\phi}(\mathbf{x}_i) \odot \epsilon^{(i)}$,
- Estimate

$$\hat{\mathcal{L}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left[\log p_{\theta}(\mathbf{x}_i | \mathbf{z}^{(i)}) - \text{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}_i) \| p(\mathbf{z})) \right],$$

- Update (θ, ϕ) by ascending the gradient of $\hat{\mathcal{L}}$.

Interpretations and Summary

1. Approximate Bayesian inference.

- ELBO decomposition:

$$\log p_{\theta}(x) = \mathcal{L}(\theta, \phi; x) + \text{KL}(q_{\phi}(z | x) \| p_{\theta}(z | x)).$$

- Maximizing ELBO $\Rightarrow$ approximate maximum likelihood and make $q_{\phi}(z | x)$ close to the true posterior.

2. Rate-distortion / information bottleneck view.

- Reconstruction term: fidelity to data.
- $\text{KL}(q_{\phi}(z | x) \| p(z))$: “rate” / complexity of the code z .

3. VAE in one sentence.

A VAE is a latent-variable generative model trained by maximizing a variational lower bound on log-likelihood, with:

- a decoder $p_{\theta}(x | z)$ (generator),
- an encoder $q_{\phi}(z | x)$ (amortized approximate posterior),
- and the reparameterization trick for low-variance gradients.

Analytical KL Divergence

For Gaussian distributions, the KL divergence has a closed form:

$$\text{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)) = \frac{1}{2} \sum_{j=1}^J (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1) \quad (1)$$

where J is the dimension of the latent space.

Interpretation:

- μ_j^2 : Penalizes large mean values
- $\sigma_j^2 - \log \sigma_j^2 - 1$: Encourages $\sigma_j \approx 1$
- Total effect: Regularizes latent space to be close to standard Gaussian

Benefits:

- No sampling required for KL term
- Stable gradients
- Interpretable regularization

Simplified Loss: Unit Variance Case

Setting $\sigma_{\text{encoder}} = 1$ and $\sigma_{\text{decoder}} = 1$:

Encoder: $q_{\phi}(z|x) = \mathcal{N}(\mu_{\phi}(x), I)$

Decoder: $p_{\theta}(x|z) = \mathcal{N}(\mu_{\theta}(z), I)$

Simplified KL term:

$$\text{KL}(q_{\phi}(z|x) \| p(z)) = \frac{1}{2} \|\mu_{\phi}(x)\|^2 \quad (2)$$

(since $\sigma_j = 1$ for all j , we have $\sigma_j^2 - \log \sigma_j^2 - 1 = 0$)

Simplified reconstruction term:

$$\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] = -\frac{1}{2} \mathbb{E}_{q_{\phi}(z|x)}[\|x - \mu_{\theta}(z)\|^2] + \text{const} \quad (3)$$

Total loss:

$$\mathcal{L}(\theta, \phi) = \frac{1}{2} \mathbb{E}_{q_{\phi}(z|x)}[\|x - \mu_{\theta}(z)\|^2] + \frac{1}{2} \|\mu_{\phi}(x)\|^2 \quad (4)$$

Interpretation:

- First term: Reconstruction error (MSE)
- Second term: Regularization on encoder mean

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β -VAE: Controlling the Regularization

Motivation: Balance between reconstruction and regularization

β -VAE objective:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \beta \text{KL}(q_{\phi}(z|x) \| p(z)) \quad (5)$$

Effect of β :

- $\beta > 1$: Stronger regularization $\Rightarrow$ Better disentanglement
- $\beta < 1$: Weaker regularization $\Rightarrow$ Better reconstruction
- $\beta = 1$: Standard VAE

Disentanglement: β -VAE encourages learning of independent latent factors

VQ-VAE: Vector Quantized VAE

Key idea: Use discrete latent representations instead of continuous
Architecture:

- Encoder: $z_e = E(x)$ (continuous)
- Quantization: $z_q = \text{Quantize}(z_e)$ (discrete)
- Decoder: $\hat{x} = D(z_q)$

Quantization process:

$$z_q = \arg \min_k \|z_e - e_k\|^2 \quad (6)$$

where $\{e_k\}_{k=1}^K$ are learnable codebook vectors.

Benefits:

- Discrete representations (useful for language, etc.)
- Better compression
- Enables autoregressive modeling in latent space

VQ-VAE Training

Challenge: Quantization is non-differentiable

Solution: Straight-through estimator

$$\frac{\partial \mathcal{L}}{\partial z_e} = \frac{\partial \mathcal{L}}{\partial z_q} \quad (7)$$

Training objective:

$$\mathcal{L} = \log p(x|z_q) + \|\text{sg}(z_e) - e_k\|^2 + \beta \|z_e - \text{sg}(e_k)\|^2 \quad (8)$$

where $\text{sg}(\cdot)$ is stop-gradient operator.

Components:

- Reconstruction loss: $\log p(x|z_q)$
- Codebook loss: $\|\text{sg}(z_e) - e_k\|^2$ (updates codebook)
- Commitment loss: $\beta \|z_e - \text{sg}(e_k)\|^2$ (encourages encoder to commit)

WAE: Wasserstein Autoencoder

Motivation: Replace KL divergence with Wasserstein distance

WAE objective:

$$\mathcal{L} = \mathbb{E}_{p(x)} \mathbb{E}_{q_\phi(z|x)} [c(x, D(z))] + \lambda \mathcal{D}(q_\phi(z), p(z)) \quad (9)$$

where:

- $c(x, D(z))$: Reconstruction cost
- $\mathcal{D}(q_\phi(z), p(z))$: Distance between aggregated posterior and prior
- λ : Regularization weight

Benefits:

- More flexible than KL divergence
- Better mode coverage
- Can use different distance metrics

VAE-GAN: Combining VAE and GAN

Idea: Use GAN discriminator to improve VAE reconstruction quality

Architecture:

- VAE encoder-decoder: $x \rightarrow z \rightarrow \hat{x}$
- GAN discriminator: Distinguishes real vs reconstructed samples
- Generator: VAE decoder

Training objective:

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \lambda \mathcal{L}_{\text{GAN}} \quad (10)$$

where:

- $\mathcal{L}_{\text{VAE}}$: Standard VAE loss
- $\mathcal{L}_{\text{GAN}}$: Adversarial loss for discriminator

Benefits:

- Better sample quality than VAE alone
- Maintains inference capabilities

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Hierarchical VAE

Motivation: Learn hierarchical latent representations

Architecture:

$$p_{\theta}(x, z_1, \dots, z_L) = p_{\theta}(x|z_1) \prod_{l=1}^{L-1} p_{\theta}(z_l|z_{l+1}) p_{\theta}(z_L) \quad (11)$$

Inference:

$$q_{\phi}(z_{1:L}|x) = \prod_{l=1}^L q_{\phi}(z_l|z_{<l}, x) \quad (12)$$

Benefits:

- Multi-scale representations
- Better modeling of complex data
- Hierarchical generation process

Conditional VAE (C-VAE)

Extension: Incorporate conditioning information

Model:

$$p_{\theta}(x, z|c) = p_{\theta}(x|z, c)p(z) \quad (13)$$

Inference:

$$q_{\phi}(z|x, c) = \mathcal{N}(z; \mu_{\phi}(x, c), \sigma_{\phi}^2(x, c)) \quad (14)$$

Applications:

- Conditional generation
- Controlled synthesis
- Multi-modal learning

Training:

$$\mathcal{L} = \mathbb{E}_{q_{\phi}(z|x, c)}[\log p_{\theta}(x|z, c)] - \text{KL}(q_{\phi}(z|x, c) \| p(z)) \quad (15)$$

Importance Weighted Autoencoder (IWAE)

Motivation: Better approximation of log-likelihood

IWAE objective:

$$\mathcal{L}_K = \mathbb{E}_{z_1, \dots, z_K \sim q_\phi(z|x)} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x, z_k)}{q_\phi(z_k|x)} \right] \quad (16)$$

Key properties:

- $\mathcal{L}_K \geq \mathcal{L}_{K-1}$ (monotonic improvement)
- $\lim_{K \rightarrow \infty} \mathcal{L}_K = \log p_\theta(x)$ (exact likelihood)
- $K = 1$ recovers standard VAE

Trade-offs:

- Better approximation with larger K
- Higher computational cost
- More complex gradients

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Common Issues and Solutions

Posterior collapse:

- Symptom: $q_{\phi}(z|x) \approx p(z)$ (no information in latent)
- Causes: Strong decoder, weak encoder
- Solutions: KL annealing, free bits, stronger encoder

Blurry reconstructions:

- Cause: Gaussian decoder assumption
- Solutions: VAE-GAN, better decoder architectures

Mode collapse:

- Symptom: Limited diversity in generated samples
- Solutions: Better regularization, architectural improvements

Evaluation Metrics

Reconstruction quality:

- MSE, SSIM, FID (Fréchet Inception Distance)
- Perceptual losses

Generation quality:

- FID, IS (Inception Score)
- Human evaluation

Representation quality:

- Disentanglement metrics (β -VAE metric, MIG)
- Downstream task performance
- Interpolation smoothness

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Summary

Key contributions of VAE:

- Scalable variational inference for latent variable models
- End-to-end training with reparameterization trick
- Foundation for many generative models

VAE variants address:

- Disentanglement (β -VAE)
- Discrete representations (VQ-VAE)
- Sample quality (VAE-GAN, WAE)
- Hierarchical modeling (HVAE)

Impact:

- Influenced modern generative models
- Enabled practical latent variable modeling
- Foundation for many applications

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. How does the choice of prior $p(z)$ affect VAE performance? What are the trade-offs between different priors?
2. Can you derive the gradient of the ELBO with respect to the encoder parameters ϕ ? How does the reparameterization trick make this tractable?
3. What are the theoretical guarantees of VAE? Under what conditions does the ELBO provide a good approximation to the true log-likelihood?
4. How would you extend VAE to handle sequential data (like in language modeling)? What are the key challenges?

Welcome inputs

Any comments?

Welcome your inputs!